

Islamic ethical frameworks regarding artificial intelligence and biotechnology

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SYSTEM PROMPT

You are an expert AI content generator specializing in creating structured article.

USER PROMPT

Write a concise, self-contained article entry for the title: "Islamic ethical frameworks regarding artificial intelligence and biotechnology".

Overall topic for alignment: Islam and technology.

- Output format: GitHub-Flavored Markdown. The markdown must be clean—do not include any code fences or surrounding markers. Target writing format: Blog Post. Follow these format-specific rules exactly:
- Aim for at least 6000 characters while staying under 12000. Prefer verifiable facts from the provided sources; do not fabricate citations or specific data.
- Place external links only in the "References" section; do not include inline external links in the body text.
- Maximum content length (including headings and references): 12000 characters. If nearing the limit, prioritize the most informative sections and keep references to 2-3.
- Write in the target response language. Avoid fluff; be precise and actionable. Do not include images or tables.
- Do not include code fences or any text outside the markdown content.

No external web sources are attached. Do not fabricate citations or specific facts that cannot be verified. Keep claims general and verifiable. In the 'References' section, include: "No external sources used."

****Respond in English****

Target audience: GENERAL. Writing tone: NEUTRAL.

SUMMARY

Summary: Islamic Ethical Frameworks Regarding Artificial Intelligence and Biotechnology

Islamic tradition approaches modern technology not through rejection, but by applying established theological principles to new realities. This framework views humans as stewards (*Khalifa*) charged with pursuing beneficial knowledge while respecting divine boundaries. The ethical evaluation of technology relies primarily on the *Maqasid al-Shari'ah* (Objectives of Islamic Law), which assesses innovations based on their ability to preserve five core values: Faith, Life, Intellect, Lineage, and Property.

Artificial Intelligence Ethics

- **Justice and Privacy:** Islamic ethics demand the elimination of algorithmic bias, viewing discrimination as a violation of divine justice (*Adl*). Strict privacy rights forbid mass surveillance and invasive data harvesting.
- **Agency and Personhood:** AI is defined strictly as a tool. Because machines lack a soul (*Ruh*), they cannot possess moral agency or legal personhood.
- **Intellect:** Technologies that augment human intelligence are encouraged, while those that deceive (deepfakes) or cause cognitive atrophy are rejected for threatening the preservation of Intellect.

Biotechnology Ethics

- **Therapy vs. Enhancement:** A critical distinction is made between "treatment" and "alteration." Genetic engineering is permitted to cure disease (restoring health) but prohibited for enhancement (altering the natural order).
- **Stem Cells and Cloning:** Stem cell research is generally permitted on embryos prior to ensoulment (typically 120 days). Therapeutic cloning for organ tissues is allowed, but reproductive cloning is banned for violating the sanctity of Lineage (*Nasl*) and family structure.

Legal Maxims (*Qawaid Fiqhiyyah*) Jurists utilize specific maxims to resolve micro-ethical dilemmas:

1. **Default Permissibility:** All technology is permitted unless proven harmful.
2. **Harm Elimination:** Technologies must undergo risk-benefit analyses; harm must be removed.
3. **Facilitating Ease:** Technology should reduce human hardship.
4. **Precautionary Principle:** The prevention of significant harm takes precedence over the acquisition of potential benefits.

Ultimately, Islamic bioethics advocates for a balanced, centrist path (*Wasat*), embracing technological progress that serves the common good without compromising human dignity or spiritual obligations.

Islamic Ethical Frameworks Regarding Artificial Intelligence and Biotechnology

The intersection of rapidly advancing technology and religious ethics presents one of the most complex intellectual challenges of the 21st century. As Artificial Intelligence (AI) and biotechnology reshape the definitions of agency, life, and intelligence, religious communities are turning to their theological foundations to navigate these uncharted waters. Within the Islamic tradition, the response to these advancements is not a rejection of modernity but a rigorous application of established jurisprudential principles to new realities.

Islam possesses a rich history of scientific inquiry, often referred to as the Islamic Golden Age, where faith and reason were seen as complementary rather than contradictory. Today, Muslim ethicists, jurists (*fuqaha*), and scientists are collaborating to construct a framework that embraces technological benefits while mitigating spiritual and social harms. This article explores the core Islamic ethical frameworks governing the development and deployment of AI and biotechnology, focusing on the concepts of stewardship (*Khilafah*), the objectives of Islamic law (*Maqasid al-Shari'ah*), and specific legal maxims (*Qawaid Fiqhiyyah*).

The Theological Foundation: Stewardship and Knowledge

To understand the Islamic approach to technology, one must first understand the theological status of the human being. In Islamic belief, humans are appointed as *Khalifa* (stewards or vicegerents) on Earth. This status confers both dignity and responsibility. Innovation is not merely permitted; it is often viewed as a communal obligation (*Fard Kifayah*) if it serves the common good. However, this stewardship implies that humans are trustees, not absolute owners, of the natural world.

Consequently, technological development is bound by the limits set by the Creator. The pursuit of knowledge (*Ilm*) is a central tenet of Islam, but it is categorized into beneficial knowledge (*Ilm Nafi*) and harmful or useless knowledge. Technologies that alleviate suffering, cure diseases, or improve quality of life are generally embraced as manifestations of divine mercy working through human intellect. Conversely, technologies that destroy the environment, undermine human dignity, or challenge the fundamental nature of creation are viewed with skepticism or prohibition.

Maqasid al-Shari'ah: The Higher Objectives

The primary lens through which contemporary Islamic bioethics and tech ethics are viewed is *Maqasid al-Shari'ah* (the higher objectives of Islamic law). These objectives serve as a compass for moral decision-

making, ensuring that rulings preserve the holistic well-being of the individual and society. There are five universally accepted objectives:

1. **Preservation of Faith (*Din*)**
2. **Preservation of Life (*Nafs*)**
3. **Preservation of Intellect (*Aql*)**
4. **Preservation of Lineage (*Nasl*)**
5. **Preservation of Property (*Mal*)**

Every technological advancement is weighed against these five pillars. If a technology threatens one of these objectives, it is deemed unethical; if it protects or enhances them, it is deemed ethical.

Artificial Intelligence through the Lens of Islamic Ethics

Artificial Intelligence introduces profound questions regarding agency, privacy, and the nature of intelligence. Islamic scholars address these through the specific application of the Maqasid and concepts of justice (*Adl*).

Algorithmic Justice and Bias

Islam places an immense emphasis on justice (*Adl*) and the prohibition of oppression (*Zulm*). In the context of AI, this translates to a rigorous critique of algorithmic bias. If an AI system used in hiring, policing, or lending discriminates against individuals based on race, gender, or background due to biased training data, it violates the Islamic mandate for justice. Developers are seen as morally accountable for the unintended consequences of their code. The maxim "Actions are judged by intentions" (*Innamal a'malu bin-niyyat*) applies, but negligence in foreseeing harm is also culpable.

Privacy and Surveillance

The right to privacy is deeply embedded in Islamic teachings. The Quran explicitly forbids spying (*Tajassus*) and entering homes without permission. In the age of Big Data, where AI systems harvest vast amounts of personal information, Islamic ethics emphasize the sanctity of private life (*Hurmah*). Data collection requires informed consent and must be limited to what is necessary. The use of AI for mass surveillance by states or corporations, which strips individuals of their dignity and autonomy, conflicts with the protection of the individual's "Nafs" (self) and "Mal" (property/data).

Automation and the Intellect

The preservation of Intellect (*Aql*) is a critical objective. While AI can augment human intelligence, scholars warn against over-reliance that leads to cognitive atrophy. Furthermore, the use of AI to generate deepfakes or spread misinformation is strictly prohibited as it corrupts the truth and deceives the intellect. The human mind is a divine gift; technologies that cloud judgment or manipulate perception are viewed as threats to *Aql*.

Robotics and Personhood

A distinct theological line is drawn regarding the nature of the soul (*Ruh*). Islam teaches that the soul is a unique divine command bestowed only upon biological life. Therefore, no matter how advanced an AI becomes, it can never possess a soul or attain true moral agency. Robots are tools, not persons. Granting legal personhood to AI, as debated in some secular jurisdictions, is generally rejected in Islamic thought because it muddies the distinction between the Creator's work (humans) and human creation (machines), and complicates accountability.

Biotechnology and the Limits of Intervention

Biotechnology, particularly genetic engineering, elicits nuanced responses from Islamic jurists, often hinging on the distinction between "treatment" (*Tadawi*) and "alteration" (*Taghyir*).

Genetic Engineering: Therapy vs. Enhancement

The majority of Islamic scholars permit genetic engineering for therapeutic purposes. If a fetus carries a gene for a debilitating disease, editing that gene to prevent suffering aligns with the preservation of Life (*Nafs*) and Lineage (*Nasl*). This is viewed as a form of medical treatment, which the Prophet Muhammad encouraged.

However, genetic engineering for "enhancement"—such as selecting for eye color, height, or intelligence—is widely rejected. This crosses the line into "altering Allah's creation" (*Taghyir Khalq Allah*), a concept derived from Quranic verses warning against Satan's promise to mislead humans into defacing the natural order. Enhancement implies a dissatisfaction with divine decree and creates social inequities that violate principles of justice.

Stem Cell Research and the Status of the Embryo

Islamic bioethics is relatively permissive regarding stem cell research compared to some other religious traditions. This stems from the Islamic understanding of ensoulment. While the embryo is respected from conception, most schools of thought believe the soul enters the body at 120 days (or 40 days in some interpretations). Therefore, research on pre-ensouled embryos (blastocysts), particularly those left over from IVF procedures that would otherwise be discarded, is often permitted if the goal is to save lives or cure diseases. This utilitarian approach prioritizes the living (patients needing cures) while maintaining respect for potential life.

Cloning

Reproductive cloning (creating a genetic copy of a human) is almost universally prohibited in Islamic jurisprudence. It is seen as a violation of the preservation of Lineage (*Nasl*) and the family structure. Islam places high importance on biological parentage and the definition of relationships. Cloning disrupts the natural order of paternity and maternity. However, *therapeutic* cloning (somatic cell nuclear

transfer) to create tissues or organs for transplant is often viewed permissibly, as it serves the preservation of Life without creating a new human entity.

Legal Maxims (Qawaid Fiqhiyyah) in Technology

Beyond the broad objectives, jurists utilize specific legal maxims to solve micro-ethical dilemmas in tech. These short, general rules facilitate decision-making when specific textual evidence is absent.

1. **"The original rule in all things is permissibility" (*Al-asl fi al-ashya al-ibahah*):** This is the starting point for all technology. AI and biotech are not forbidden by default. The burden of proof lies on those who wish to prohibit a technology to show that it causes harm or violates a tenet of faith.
2. **"Harm must be eliminated" (*Ad-darar yuzal*):** This is perhaps the most frequently cited maxim in bioethics. If a medical technology causes side effects that outweigh the benefits, it is prohibited. In AI, if an autonomous vehicle algorithm prioritizes speed over safety, it violates this maxim. This principle essentially mandates a risk-benefit analysis for every new innovation.
3. **"Hardship brings facility" (*Al-mashaqqah tajlib at-taysir*):** This maxim supports the use of technology to ease human burdens. Robotic surgery that reduces recovery time, or AI that helps the visually impaired navigate the world, are championed under this rule. It underscores that Islam seeks to remove unnecessary difficulty from human life.
4. **"Prevention of harm takes precedence over the acquisition of benefits" (*Dar' al-mafasid muqaddam 'ala jalb al-masalih*):** When a technology has both significant risks and significant rewards (e.g., gain-of-function research in virology or unconstrained Artificial General Intelligence), this maxim suggests a precautionary approach. If the potential catastrophic harm is probable, the technology should be restricted even if it offers benefits.

Conclusion

The Islamic ethical framework regarding artificial intelligence and biotechnology is neither luddite nor radically accelerationist. It is a balanced, centrist path (*Wasat*) that seeks to harness the power of human ingenuity within the boundaries of divine ethics. By leveraging the objectives of *Maqasid al-Shari'ah*, the theological concept of stewardship, and flexible legal maxims, Islamic thought offers a robust mechanism for evaluating modern science.

The consensus among Muslim scholars is that technology is a tool—neutral in its essence but moral in its application. The ultimate goal of integrating these technologies is not merely economic growth or efficiency, but the facilitation of a just society that honors the dignity of the human being and the sanctity of the Creator's design. As AI and biotech continue to evolve, the dialogue between Islamic scholars and the scientific community remains vital to ensure that humanity remains the master of its tools, rather than a victim of its own inventions.

SEGMENTED DATA (2)

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References

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